Q1. This question is about four chemical elements.

(a) The melting points and boiling points of the four elements are shown in the table. Complete the table to give the physical state, **solid**, **liquid** or **gas**, of each element at room temperature, 21°C.

element	melting point in °C	boiling point in °C	physical state at room temperature, 21°C
bromine	–7	59	
chlorine	–101	–34	
fluorine	–220	–188	
iodine	114	184	

4 marks

(b) Bromine can be a solid, a liquid or a gas depending on the temperature.
In which physical state will 10 g of bromine store the most thermal energy?

.....

1 mark

(c) Is bromine a **solid**, a **liquid** or a **gas** when the arrangement of particles is:

(i) far apart and random? .....

1 mark

(ii) close together but random? .....

1 mark

(iii) close together in a regular pattern? .....

1 mark

Maximum 8 marks

Mark schemes

Q1.

(a)

bromine	liquid
chlorine	gas
fluoride	gas
iodine	solid

4 (L6)

(b) gas

1 (L6)

(c) (i) gas

1 (L6)

(ii) liquid

1 (L6)

(iii) solid

1 (L6)

[8]